

Mohit Appari

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SUMMARY

AI/ML Engineer with **3+ years of experience** building scalable **machine learning, GenAI, forecasting, and data-intensive systems** across research and production environments. Skilled in **time-series forecasting, statistical modeling, predictive analytics, hypothesis testing, ML inference, and quantitative analysis**, with experience processing **millions of records and 10K+ financial documents daily**. Delivered measurable impact including **83% reduction** in SQL/report execution time, **80% reduction** in software licensing costs, and **15% reduction** in ML validation error across **11K+ material structures**. Bridges **AI engineering, data science, and quantitative research** to build scalable solutions for financial markets and FinTech.

EXPERIENCE

Agency of Persons with Disabilities - State of Florida

Data Scientist

Jun 2025 – Present

- Analyzed **millions of transactional records** across **60K+ clients and \$200M+ in monthly spending**, performing exploratory and statistical analysis to uncover **anomalies, utilization trends, data inconsistencies, and operational patterns** that informed financial and service decisions.
- Developed **spending and budget forecasting models** using **ARIMA, regression, moving averages, and time-series decomposition**, generating projections typically within **5–10% of actual expenditures** to support budgeting, rate decisions, and strategic planning.
- Modeled reimbursement-rate scenarios across **22+ service areas**, quantifying projected financial impact under alternative rates and translating historical spending and utilization patterns into **data-driven recommendations** for leadership.
- Engineered **40+ automated ETL and analytics pipelines** using **Python, SQL Server, and Azure Data Warehouse**, processing **300K+ records** and enabling scalable reporting, forecasting, anomaly analysis, and data-driven workflows.
- Re-engineered statistical and reporting workflows using **Python, SQL, SPSS, and Power BI**, reducing analysis turnaround time by **75%** and SQL processing time by **83%**, while delivering stakeholder-facing dashboards and ad hoc analyses that turned complex data into actionable insights.

Florida State University

AI/ ML Researcher

June 2024 – Sept 2024

- Fine-tuned and benchmarked **NequIP**, an equivariant neural network, using **transfer learning across 11K+ material structures**, optimizing hyperparameters and custom **MAE/MSE weighted loss functions** to improve prediction consistency and reduce validation error by **~15%**.
- Built reproducible **Python, Docker, MLflow, and HPC** workflows for data preparation, model training, experiment tracking, inference, and evaluation, incorporating **Neo4j** to represent and analyze relationships within structured scientific data, which streamlined experiment management and enabled faster, repeatable model testing.
- Designed controlled experiments across multiple model architectures using **statistical hypothesis testing, A/B comparisons, and out-of-sample validation**, to identify best architecture which improved subsequent models.

SP Global

Software Developer

Jan 2022 – Aug 2023

- Led the modernization of legacy .NET document-processing workflows by researching and evaluating open-source **Python OCR** and document-processing technologies, translating existing requirements into a scalable Python extraction platform and reducing licensing costs by **80%**.
- Mentored a 5-person engineering team** on Python-based data extraction, pipeline development, and implementation practices while contributing to the architecture and production rollout of systems processing **10,000+ financial, energy, and automotive documents/day**, achieving **95% extraction accuracy** across the broader workflow.
- Engineered fault-tolerant **AWS/Docker data pipelines** with event-driven ingestion, automated retries, structured logging, validation, and server-based automated processing, reducing manual extraction effort by **90%+** and enabling unattended data capture.
- Built OCR and CNN-based classification workflows using Tesseract and TensorFlow to transform unstructured financial filings, earnings information, energy specifications, and automotive documents into structured datasets, enabling downstream analytics and reducing manual data-entry effort.
- Designed modular Python frameworks and cloud data workflows using **AWS S3, Athena, BigQuery, REST APIs, CDC-based updates, GitHub Actions CI/CD**, and **80% test coverage**, improving scalability, data quality, deployment reliability, and maintainability across business domains.

LG Electronics

Software Developer Intern

May 2021 – Aug 2021

- Developed real-time **drowsiness detection and logistics tracking systems** using Python, OpenCV, YOLO, CNNs, REST APIs, and PostgreSQL, achieving **90% detection accuracy**, reducing sorting time by **25%**, and earning **3rd place in LG's internal Ideathon**.

ITC Infotech

Software Developer Intern

Jan 2021 – Mar 2021

- Developed **Python Flask REST APIs** with MySQL for master-data management, role-based access control, and automated product tracking workflows, reducing **manual workload by 30%**.

EDUCATION

Florida State University | Tallahassee, FL

Aug 2023 – May 2025

Masters of Science in Data Science, GPA 4/4

SKILLS

AI/ML: LLMs, RAG, MCP, AI Agents, LangChain, LangGraph, LlamaIndex, Codex, Claude, Gemini, Cohere, Ollama, Supabase, Pinecone, W&B
Quantitative Finance: Market Research, Equity Analysis, Derivatives, Options Pricing, Greeks, Volatility, Financial Modeling, Alpha Research
Data Engineering & Infrastructure: PostgreSQL, MySQL, Oracle, MongoDB, Redis, Cassandra, Neo4j, DuckDB, Snowflake, Delta Lake, AWS, GCP, Azure, Kubernetes, Docker, Terraform, Apache Airflow, dbt, Kafka, Hadoop, Spark, Git, GitHub Actions, CI/CD, Linux, Bash